AC2206 & AC3302

Investment Appraisal

Autumn 2021 Examination

Paper 1 · 1.5 hours · Answer TWO Questions only · Each question worth 50 Marks

Module	AC2206 & AC3302 — Investment Appraisal
Session	Autumn 2021
Paper	1 of 1
External Examiner	Professor Edward Jones
Internal Examiner	Professor Raymond Donnelly
Structure	**Different from other papers in this series:** there is no compulsory Question 1 here — answer any TWO of the three questions, each worth 50 Marks. This is also an open-book exam.
This scheme covers	All three questions in full, so you can choose your strongest pair going into the exam

Own-figure rule. Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step.

Question 1 involves genuine, unavoidable timing ambiguity. Unlike the tightly-specified NPV questions in other papers in this series, Question 1 describes a project with a staged setup phase and does not pin down every cash flow's exact timing with full precision. This scheme states every timing assumption explicitly as it goes, picks one internally-consistent interpretation, and flags clearly where an equally defensible alternative exists — see the checknotes throughout Question 1(a) in particular.

Quality check. Every calculation in this scheme has been independently verified and cross-checked for accuracy before being written up.

Question 1

50 Marks — Full Project NPV with Grants, Staged Investment & Relevant Costs

Scenario

Cepheus Limited is considering extending its factory to manufacture office seating. Key facts: €10,000 already paid to architects for planning permission (a **sunk cost**); a further €30,000 payable to architects to manage construction; construction cost €2 million (€500,000 paid immediately, balance of €1.5m on completion in 1 year); machinery €500,000, ordered now, paid when installed (on completion, 1 year from now); a 25% grant on the extension AND machinery cost, paid 1 year after each expenditure is incurred; 8 new-recruit operatives at €30,000/year each; 3 operatives transferred internally (no replacement needed, so no incremental cost for 2 of them) with one promoted to production manager at €50,000/year (up from €30,000); fixed production costs €100,000/year; contribution margin €300/unit on 3,000 units/year; advertising €50,000 (year 1), €25,000 (years 2–4), €20,000 (year 5 onward); working capital €200,000, required only once production commences; a 5-year horizon (NPV of cash flows after year 5 is zero). Cost of capital: **6%**. No tax is mentioned in this question.

Required

50 Marks

Should Cepheus proceed to produce the new product? Show your workings.

FULL MODEL ANSWER

Step 1 — Identify relevant vs irrelevant cash flows

> **The €10,000 already paid to architects is a sunk cost:** It has been incurred regardless of whether the project proceeds and must be EXCLUDED from the NPV calculation — including it would violate the fundamental relevant-cost principle of investment appraisal.

> **Only 1 of the 3 transferred operatives creates an incremental cost:** Two of the three transferred operatives continue to be paid their existing €30,000 salary, which the company would have paid regardless (since automation means their old roles do not need replacing) — moving them to the new facility creates no incremental cash flow. The third, promoted to production manager, moves from €30,000 to €50,000 — only the €20,000 increase is incremental, since the €30,000 base was already a committed cost.

Step 2 — Timing assumptions used in this scheme (stated explicitly)

This scheme uses the following timeline: **t = 0** (now) — decision point, initial architect fee and first builder payment. **t = 1** (one year from now) — extension complete, machinery installed and paid, staff transfer happens, working capital committed, production begins. Each subsequent **production year's** net operating cash flow (contribution less fixed costs, staff costs and advertising) is recognised at the END of that year, i.e. production Year 1's cash flow lands at **t = 2**, Year 2 at **t = 3**, ... Year 5 at **t = 6**. Grants are paid exactly 1 year after the specific expenditure that triggered them, applied to each tranche of extension/machinery spend separately. **This is one reasonable, internally-consistent reading of a genuinely ambiguous timeline** — the question does not pin down whether "year 1 of production" cash flows should land at t=1 or t=2. A candidate who instead compressed everything by one year (production Year 1 landing at t=1) has used an equally defensible convention and should receive full method marks for a consistently-applied alternative timeline (own-figure).

Question 1 (continued)

FULL MODEL ANSWER

Step 3 — Grants

Expenditure	Timing	Amount	Grant (25%)	Grant Received
Extension — builder deposit	t = 0	€500,000	€125,000	t = 1
Extension balance + machinery	t = 1	€1,500,000 + €500,000 = €2,000,000	€500,000	t = 2

Step 4 — Annual operating cash flow during production (Years 1-5)

Item	Amount per year
Contribution (3,000 units × €300)	€900,000
Less: Fixed production costs	(€100,000)
Less: Incremental staff cost (8 × €30,000 new recruits + €20,000 manager increment)	(€260,000)
Less: Advertising (varies by year — see below)	(varies)

Production Year	Advertising	Net Operating Cash Flow
1	€50,000	**€490,000**
2	€25,000	**€515,000**
3	€25,000	**€515,000**
4	€25,000	**€515,000**
5	€20,000	**€520,000**

Question 1 (continued)

FULL MODEL ANSWER

Step 5 — Full cash flow timeline and NPV

t	Cash Flow Items	Net Cash Flow	PV Factor @ 6%	Present Value
0	Architect fee (−€30,000) + builder deposit (−€500,000)	−€530,000	1.000	−€530,000
1	Builder balance (−€1,500,000) + machinery (−€500,000) + working capital (−€200,000) + grant (+€125,000)	−€2,075,000	0.943	−€1,956,725
2	Year 1 operating CF (+€490,000) + grant (+€500,000)	+€990,000	0.890	+€881,100
3	Year 2 operating CF	+€515,000	0.840	+€432,600
4	Year 3 operating CF	+€515,000	0.792	+€407,880
5	Year 4 operating CF	+€515,000	0.747	+€384,705
6	Year 5 operating CF (+€520,000) + working capital recovery (+€200,000)	+€720,000	0.705	+€507,600
NPV				**+€127,160**

Recommendation: Cepheus should proceed. The project has a positive NPV of approximately **+€127,160** at the 6% cost of capital, indicating it is expected to create shareholder value — though, given the scale of the investment (over €2.5m gross), the margin is not large, so the recommendation is more sensitive to the underlying assumptions (particularly the advertising/staffing cost estimates and the timing convention flagged above) than a typical NPV problem.

Working capital recovery, flagged as an assumption. As with similar questions elsewhere in this series, the paper does not explicitly state whether the €200,000 working capital is recovered at the end of the project. This scheme assumes full recovery at t=6 (the standard convention once the project's competitive advantage is expected to erode and cash flows beyond that point net to zero) — a candidate who instead treated it as unrecovered would get a correspondingly lower, but still creditable, own-figure NPV.

Mark Allocation — Question 1, 50 Marks

Tier	Marks	What separates this tier
40–50	40–50	Sunk cost correctly excluded; incremental (not total) staff cost correctly identified; grants correctly timed and calculated; full, internally consistent cash flow timeline constructed and clearly stated; correct final NPV with a clear recommendation.
28–39	28–39	Correct overall structure with one or two component errors — e.g. sunk cost included, or grant timing/amount miscalculated, or staff cost not correctly reduced to the incremental amount (own-figure applied thereafter, provided the approach is internally consistent).
15–27	15–27	Core operating cash flow (contribution less costs) correctly identified but grants, staged investment timing, or working capital not correctly incorporated.
0–14	0–14	No credible attempt to build a relevant-cash-flow NPV, or the sunk cost and full (non-incremental) staff costs both included without adjustment.

Question 1 Total: 50 Marks

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

How to use this scheme. This paper requires any TWO of the three questions, each worth 50 marks and weighted equally. This scheme covers all three in full so you can choose your strongest pair.

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

(a) Required

5 Marks

Jolanda invested €100 in a bank on 1st January 2017. If her account pays 2% per annum how much is in her account four years later on 31st December 2020? Show your workings.

FULL MODEL ANSWER

$$FV = PV \times (1+r)^n = €100 \times (1.02)^4$$

Component	Value
Principal (PV)	€100.00
Rate (r)	2%
Years (1 Jan 2017 to 31 Dec 2020)	4
Future Value	**€108.24**

Mark Allocation — Part (a), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct FV with formula and working shown.
3–4	3–4	Correct formula, minor arithmetic slip.
1–2	1–2	Wrong number of periods or rate applied.
0	0	No credible attempt.

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

(b) Required

5 Marks

If the account paid the same interest rate semi-annually how much would be in Jolanda's account on 31st December 2020? Show your workings.

FULL MODEL ANSWER

A 2% nominal annual rate paid semi-annually means 1% per half-year, compounded over 8 half-year periods (4 years):

$$\mathbf{FV = €100 \times (1.01)^8}$$

Component	Value
Semi-annual rate	1%
Number of periods (4 years × 2)	8
Future Value	**€108.29**

More frequent compounding (semi-annual vs annual) at the same nominal rate produces a slightly higher future value (€108.29 vs €108.24) — the standard result that increasing compounding frequency raises the effective annual rate above the nominal rate.

Mark Allocation — Part (b), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct FV using 1% semi-annual rate over 8 periods, with working shown.
3–4	3–4	Correct approach, minor arithmetic slip.
1–2	1–2	Rate or periods not correctly adjusted for semi-annual compounding (e.g. still used 4 periods).
0	0	No credible attempt.

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

(c) Required

5 Marks

Tamarind has just paid a dividend of €1. It expects dividends to grow at 3% per annum forever. The price of a share in Tamarind is €30. What is Tamarind's cost of capital? Show your workings.

FULL MODEL ANSWER

Using the Gordon Growth Model, rearranged to solve for the cost of capital:

$$P = D_1 \div (r - g) \quad \Rightarrow \quad r = D_1 \div P + g$$

Component	Value
D_0 (just paid)	€1.00
$D_1 = D_0 \times 1.03$	€1.03
Current price (P)	€30.00
Dividend yield = $D_1 \div P$	3.43%
Growth rate (g)	3.00%
Cost of capital (r)	**6.43%**

A common error is to use D0 (€1.00) instead of D1 (€1.03) in the numerator — the Gordon Growth Model always requires next year's dividend, since P0 is the value of all dividends starting from D1.

Mark Allocation — Part (c), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct use of D1 (not D0) in the rearranged Gordon Growth formula, correct final answer.
3-4	3–4	Correct formula but D0 used instead of D1 (own-figure, gives r=6.33%).
1-2	1–2	Formula recalled but rearranged incorrectly.
0	0	No credible attempt.

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

(d) Required

5 Marks

A government bond has a face value of €100 and a coupon rate of 4%. It has 5 full years to maturity and its yield is 2%. Interest is paid annually at the end of the year and the bond will be redeemed in one fell swoop 5 years from now. What is the value of the bond today? Show your workings.

FULL MODEL ANSWER

$$\text{Value} = C \times [1 - (1+y)^{-n}] / y + FV / (1+y)^{n}$$

Component	Value
Annual coupon (4% × €100)	€4.00
PV of Annuity Factor, 5yr @ 2%	4.713
PV of coupons (4.00 × 4.713)	€18.85
PV Factor, 5yr @ 2%	0.906
PV of face value (100 × 0.906)	€90.60
Bond Value Today	**€109.45**

The bond trades at a premium to face value (€109.45 vs €100), as expected since its 4% coupon exceeds the 2% yield to maturity.

Mark Allocation — Part (d), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct bond value with coupon annuity PV and face value PV both shown and summed correctly.
3–4	3–4	Correct structure, minor arithmetic or factor-lookup error.
1–2	1–2	One component (coupon or face value) correct in isolation but not combined correctly.
0	0	No credible attempt.

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

(e) Required

15 Marks

Analysts have forecasted that next year's dividend per share for Durbin plc will be 120 cent. They also forecast that the short-run growth rate in dividends will be 4% per annum until the end of year 4. From year 5 dividends are anticipated to grow at 1% per annum indefinitely. The cost of capital of Durbin is 6%. What is the value of a share in Durbin? Show your workings.

FULL MODEL ANSWER

This is a two-stage dividend discount model: D_1 = 120c is already given as next year's dividend, growing at 4% through year 4, then settling into a constant 1% growth perpetuity from year 5 onward.

Year	Growth applied	Dividend (cent)
1 (given)	—	120.00
2	$D_1 \times 1.04$	124.80
3	$D_2 \times 1.04$	129.79
4	$D_3 \times 1.04$	134.98
5 onward	$D_4 \times 1.01$, growing 1% forever	$D_5 = 136.33$

Terminal value at end of year 4 (value of the year-5-onward growing perpetuity, discounted back to t=4):

$$\mathbf{TV_4 = D_5 \div (r - g) = 136.33 \div (0.06 - 0.01) = 2{,}726.67c}$$

Component	Present Value (cent)
PV(D1) = 120.00 ÷ 1.06	113.21
PV(D2) = 124.80 ÷ 1.06^2	111.09
PV(D3) = 129.79 ÷ 1.06^3	108.99
PV(D4) = 134.98 ÷ 1.06^4	106.94
PV(TV4) = 2,726.67 ÷ 1.06^4	2,159.72
Value per share	**2,600.0c = €26.00**

Sense check. The valuation resolves to a clean €26.00 per share — a good sign that the two-stage structure has been set up correctly (D1 taken as already given rather than requiring a further year of growth, and the terminal value correctly discounted back exactly 4 years, matching the point at which the perpetuity begins).

Mark Allocation — Part (e), 15 Marks

Tier	Marks	What separates this tier
12–15	12–15	Correct two-stage DDM structure — D1 taken as given (not grown further), dividends 2–4 grown at 4%, correct terminal value at year 4 discounted back 4 years — with correct final share value.
8–11	8–11	Correct structure with a minor arithmetic slip in one component (own-figure), e.g. terminal value discounted back the wrong number of years.
4–7	4–7	Growth stages confused (e.g. D1 grown by a further year, or short-run/long-run growth periods mixed up), but method otherwise recognisable as a two-stage DDM.
0–3	0–3	Single-stage (Gordon growth) model used throughout with no attempt to separate the two growth phases, or no credible attempt.

Question 2

50 Marks — Time Value of Money, Valuation & Loan Amortisation

(f) Required

15 Marks

Dandalla borrows €10,000 from her bank at 5% per annum over 4 years to purchase a new car. She is required to repay this amount in equal annual instalments on the anniversary of the drawdown of the loan. Draw up the full repayment schedule for Dandalla's loan.

FULL MODEL ANSWER

The equal annual instalment (annuity payment) is found by dividing the loan principal by the 4-year, 5% present value of annuity factor. Using the more precise, unrounded annuity factor (3.5460) gives the cleanest reconciliation to a zero closing balance; using the 3-decimal table factor (3.546) gives a materially identical payment with a few cents of rounding residual in year 4.

Payment = €10,000 ÷ 3.5460 (4yr annuity factor @ 5%) = €2,820.12

Year	Opening Balance	Interest @ 5%	Instalment	Principal Repaid	Closing Balance
1	€10,000.00	€500.00	€2,820.12	€2,320.12	€7,679.88
2	€7,679.88	€383.99	€2,820.12	€2,436.13	€5,243.75
3	€5,243.75	€262.19	€2,820.12	€2,557.93	€2,685.82
4	€2,685.82	€134.29	€2,820.12	€2,685.83	€0.00
Total		**€1,280.47**	**€11,280.47**	**€10,000.00**	

Reconciliation checks. (1) The closing balance in year 4 is exactly €0.00 — confirming the equal instalment fully amortises the loan over its 4-year term. (2) Total principal repaid across all four years sums to exactly €10,000.00, matching the original loan. (3) Interest declines and principal repaid rises each year, as expected for a standard reducing-balance amortising loan — a candidate whose schedule does not exhibit this pattern, or does not reach exactly zero in year 4, has an arithmetic error somewhere in the schedule. A candidate using the rounded table annuity factor (3.546 instead of 3.5460) may see a few cents of residual balance in year 4 — this is a rounding artefact, not an error, and should not be penalised.

Mark Allocation — Part (f), 15 Marks

Tier	Marks	What separates this tier
12–15	12–15	Correct annual instalment calculated from the annuity factor; full 4-year schedule correctly showing interest, principal, and closing balance each year, with closing balance reaching exactly zero in year 4.
8–11	8–11	Correct instalment and approach, but a carried-through arithmetic error in one year (own-figure, provided the balance still reduces to zero or close to it by year 4).
4–7	4–7	Instalment calculated correctly but the interest/principal split not correctly separated each year (e.g. straight-line principal repayment used instead of a reducing-balance schedule).
0–3	0–3	Instalment amount incorrect (wrong annuity factor or formula), or no credible schedule constructed.

Question 2 Total: 50 Marks

Question 3

50 Marks — Real vs Nominal Rates, Options & the NPV/IRR Conflict

(i) Required

4 Marks

The rate of inflation is 2.5%. Mythical plc has a nominal cost of capital of 5.58% and is considering a project which costs €10 million and pays annual real cash inflows of €4 million over three years commencing one year from now. Should Mythical take this project? Show your workings.

FULL MODEL ANSWER

Since the cash inflows are given in **real** terms, they must be discounted at the **real** discount rate (not the nominal rate) — mixing a real cash flow with a nominal discount rate would be an inconsistent "apples and oranges" comparison. The real rate is found using the Fisher equation:

$$(1 + \text{nominal}) = (1 + \text{real})(1 + \text{inflation}) \quad \Rightarrow \quad \text{real} = (1+\text{nominal})/(1+\text{inflation}) - 1$$

Component	Value
Nominal cost of capital	5.58%
Inflation rate	2.5%
Real rate = 1.0558/1.025 − 1	**≈ 3.00%**

Component	Value
Real annual cash inflow	€4.000m
PV of Annuity Factor, 3yr @ 3%	2.8286
PV of inflows (4.000 × 2.8286)	€11.314m
Less: Initial cost	(€10.000m)
NPV	**+€1.31m**

Recommendation: Mythical plc should take the project. The NPV is positive at approximately **+€1.31 million**, whether the real rate is taken as the precisely-derived 3.005% or rounded to a clean 3% (the figures used above) — the nominal cost of capital was evidently designed to produce almost exactly a 3% real rate given 2.5% inflation, a useful internal check that the real-rate approach has been applied correctly.

Mark Allocation — Part (i), 4 Marks

Tier	Marks	What separates this tier
4	4	Correct use of the Fisher equation to derive the real rate, correctly applied to the real cash flows, with correct NPV and recommendation.
2–3	2–3	Real rate correctly derived but a minor arithmetic slip in the NPV, or real rate approximated (nominal − inflation) rather than the full Fisher equation (own-figure).
1	1	Nominal rate incorrectly applied directly to the real cash flows without any Fisher adjustment.
0	0	No credible attempt.

Question 3

50 Marks — Real vs Nominal Rates, Options & the NPV/IRR Conflict

(ii) Required

6 Marks

Janey bought a call option on the stock of Chipstead 6 months ago for 15 cent when Chipstead was trading at 90c. The exercise price of the option is 90 cent. The current share price is 125 cent. The option is due to expire tomorrow. Do you expect Janey to profit from her purchase? Outline and carefully illustrate the profit profile of Janey's option.

FULL MODEL ANSWER

Because the option expires tomorrow, there is essentially no time value remaining — its value is effectively just its intrinsic value at the current share price:

Profit = Max(Share Price − Exercise Price, 0) − Premium Paid

Component	Value
Exercise (strike) price	90c
Current share price	125c
Intrinsic value = Max(125 − 90, 0)	35c
Less: Premium originally paid	15c
Profit	**+20c per share**

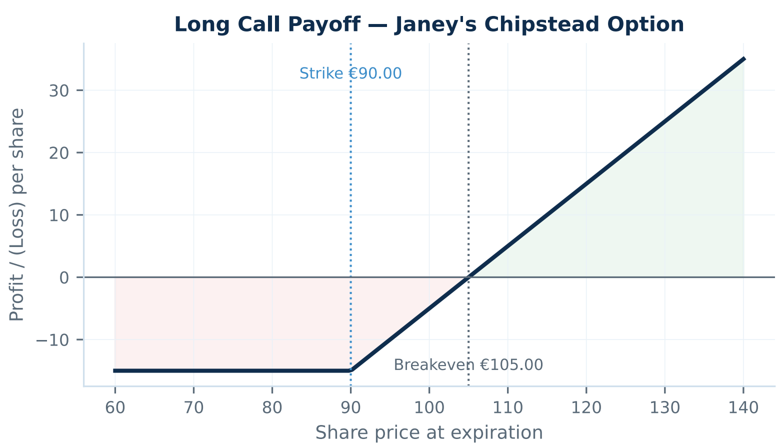

Fig. 1 — Long call payoff, Chipstead (strike 90c, premium 15c, breakeven 105c)

Yes, Janey should profit — by 20c per share. The option was bought at-the-money (strike = spot price of 90c when purchased), and the share price has since risen well above the breakeven of 105c (strike + premium = 90 + 15). With the option expiring tomorrow, Janey should exercise it (or sell it, since its market value should closely track the 35c intrinsic value) to realise this profit.

Mark Allocation — Part (ii), 6 Marks

Tier	Marks	What separates this tier
5–6	5–6	Correctly recognises that near-expiration, value ≈ intrinsic value; correct 20c profit figure; correctly labelled payoff diagram showing the kink at the 90c strike and the 105c breakeven.
3–4	3–4	Correct profit figure but diagram missing or not clearly labelled with strike/breakeven.
1–2	1–2	Profit direction correct (Janey profits) but the 20c figure incorrect.
0	0	No credible attempt, or concludes Janey does not profit.

Question 3

50 Marks — Real vs Nominal Rates, Options & the NPV/IRR Conflict

(iii) Required **10 Marks**

Fred sold Janey the at-the-money call option. He did this because he already had invested in Chipstead's stock. Explain why Fred might do this and illustrate his strategy.

FULL MODEL ANSWER

Fred's strategy is a covered call: By writing (selling) a call option against shares he already owns, Fred is running a covered call strategy — 'covered' because if the option is exercised against him, he can deliver the shares he already holds rather than having to buy them at an unknown, potentially much higher, market price (which would be the much riskier 'naked call' position).

Why Fred might do this — income generation: Fred immediately receives the 15c option premium, which enhances his overall return on the shares he holds, particularly useful if Fred expects the share price to stay relatively flat or rise only modestly over the option's life — the premium is 'found money' in that scenario.

Why Fred might do this — a partial cushion against a price fall: The premium received provides a small buffer against a decline in the share price: Fred's combined position (shares plus premium received) only starts losing money once the share price falls below his purchase cost by more than the 15c premium collected, compared to an unhedged position which starts losing immediately.

The trade-off — capped upside: In exchange for the premium income, Fred gives up any share price appreciation above the 90c strike price: if Chipstead rises above 90c (as it in fact did, to 125c), Fred is obligated to sell his shares at only 90c if the option is exercised, missing out on the additional 35c of gain that Janey (the option holder) receives instead.

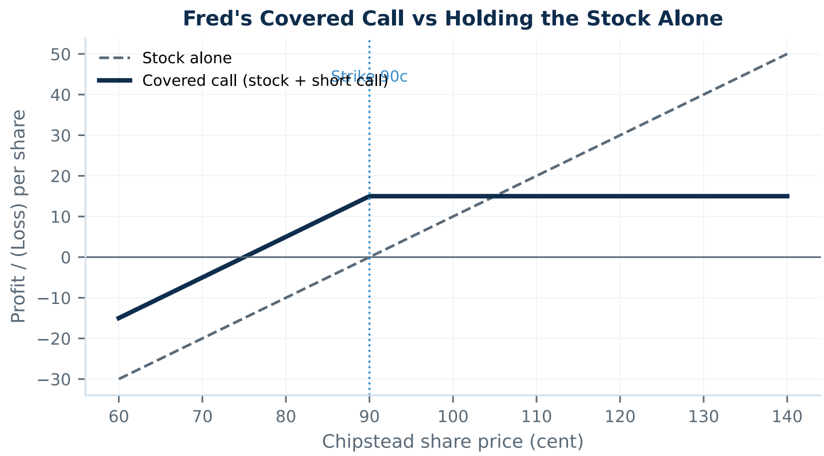

Fig. 2 — Covered call: premium income cushions modest declines, but upside is capped once the share price passes the 90c strike

What actually happened to Fred. With Chipstead now at 125c, Fred's covered call strategy has capped his gain — instead of participating in the rise from 90c to 125c (a 35c gain), he receives only the 90c strike price if exercised, plus the 15c premium already collected, for a total of 105c effectively realised per share — 20c less than if he had simply held the shares unhedged. This illustrates the central trade-off of a covered call: reliable extra income in flat or modestly rising markets, at the cost of forgoing large gains in a strongly rising market.

Mark Allocation — Part (iii), 10 Marks

Tier	Marks	What separates this tier
8–10	8–10	Correctly identifies the strategy as a covered call, with 'covered' correctly explained; explains BOTH the income-generation motive AND the capped-upside trade-off; diagram correctly shows the combined payoff shape.
5–7	5–7	Correctly identifies covered call and explains the income motive, but the capped-upside trade-off underdeveloped or the illustration missing/unclear.
2–4	2–4	Recognises Fred already owns the stock but does not clearly connect this to why he would sell a call against it.
0–1	0–1	No meaningful explanation of the covered call strategy.

Question 3

50 Marks — Real vs Nominal Rates, Options & the NPV/IRR Conflict

(iv) Cavallino Ltd — Required

30 Marks

Cavallino Ltd. is considering the following mutually exclusive projects (cost of capital 8%):

	Cost	Year 1	Year 2	Year 3
Project A	−100,000	9,641	65,000	99,000
Project B	−100,000	90,237	50,000	10,000

(i) Which project would Cavallino take based on the IRR criterion? (10 Marks)
(ii) Which project will Cavallino take on the basis of NPV? (10 Marks)
(iii) Write a short report explaining any differences between the NPV and IRR methods in this instance and more generally. (10 Marks)

FULL MODEL ANSWER

(i) IRR criterion

Project	IRR
A	**25.0%**
B	**33.4%**

On the IRR criterion, **Cavallino would select Project B**, since its IRR (33.4%) exceeds Project A's IRR (25.0%).

(ii) NPV criterion (at 8% cost of capital)

Project	PV of Inflows @ 8%	Cost	NPV
A	€143,239	€100,000	**+€43,239**
B	€134,349	€100,000	**+€34,349**

On the NPV criterion — the theoretically correct decision rule for choosing between mutually exclusive projects — **Cavallino should select Project A**, since it has the higher NPV (€43,239 vs €34,349) at the actual 8% cost of capital.

This is a genuine NPV/IRR conflict. The two criteria disagree: IRR ranks B above A, but NPV ranks A above B. This is verified as a real conflict (not a calculation error) by finding the **crossover rate** — the discount rate at which the two projects' NPVs are equal, found as the IRR of the incremental cash flow stream (B minus A): **≈14.8%**. Because Cavallino's actual cost of capital (8%) is below this crossover rate, Project A correctly has the higher NPV despite B's higher IRR — the two criteria will agree only at discount rates above 14.8%, where B's earlier, larger cash flows begin to dominate.

Question 3 (continued)

FULL MODEL ANSWER

(iii) Short report — explaining the NPV/IRR conflict

The specific cause here — differing cash flow timing (project scale over time): Project B is heavily front-loaded (€90,237 in year 1 alone, 90% of its cost recovered almost immediately), while Project A is back-loaded (the bulk of its return, €99,000, arrives only in year 3). IRR, expressed as a single percentage, tends to favour the project that pays back fastest, since a rapid return compounds more favourably under IRR's own (reinvestment) logic — even though, at the company's actual cost of capital, the more patient Project A ultimately creates more absolute value.

The reinvestment rate assumption is the underlying theoretical driver: IRR implicitly assumes that all interim cash flows are reinvested at the project's own IRR (25.0% for A, 33.4% for B) — an unrealistically high and, worse, inconsistent assumption across the two projects being compared. NPV, by contrast, assumes reinvestment at the company's actual cost of capital (8%), which is both more realistic and applied consistently to both projects, making NPV the theoretically sound basis for comparison.

Why this matters specifically for mutually exclusive projects: When choosing between two mutually exclusive projects (as here — Cavallino can only take one), the decision-relevant question is which project adds more absolute value to the firm, not which has the higher percentage return on its own money. NPV directly answers 'how much wealth does this create', while IRR answers 'what is the percentage return', and a smaller project (or a differently-timed one) can have a higher percentage return while still creating less total value — exactly the situation here, since both projects have identical initial cost, so the conflict is driven purely by the timing of the cash inflows.

General recommendation: For mutually exclusive project choices, NPV should be used as the primary decision criterion whenever NPV and IRR conflict, precisely because it directly measures value creation at the firm's actual cost of capital. IRR remains a useful supplementary metric (e.g. for communicating a project's percentage return, or for capital-constrained ranking of independent, divisible projects via the related profitability index), but should not override NPV when the two rankings disagree for mutually exclusive alternatives.

Candidates who correctly identify BOTH the proximate cause (differing timing/front-loading of cash flows between the two projects) AND the underlying theoretical driver (differing, project-specific reinvestment rate assumptions under IRR vs the consistent cost-of-capital assumption under NPV), and correctly conclude that NPV should govern the mutually-exclusive choice, should receive full credit.

Mark Allocation — Part (iv), 30 Marks (10 + 10 + 10)

Tier	Marks	What separates this tier
i: 8–10	8–10	Both IRRs correctly calculated (or reasonably estimated via interpolation) with correct project selection under the IRR criterion.
ii: 8–10	8–10	Both NPVs correctly calculated at 8% with correct project selection under the NPV criterion.
iii: 8–10	8–10	Report correctly identifies the cash flow timing difference as the proximate cause, explains the differing reinvestment rate assumption as the underlying theoretical driver, and correctly recommends NPV as the governing criterion for mutually exclusive projects.
Overall partial credit	varies	A candidate who gets (i) and (ii) right but cannot explain the conflict in (iii), or vice versa, should be marked on each part independently per the bands above — this is a three-part question with three separable skills being tested.

Question 3 Total: 50 Marks